INCH-POUND

MIL-A-70833A (AR)
03 September 1991
SUPERSEDING
MIL-A-70833 (AR)
15 April 1990

MILITARY SPECIFICATION

AUTOMATIC TANK TARGET: 11784501

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements and tests for the automatic tank target assembly of the Target Holding Mechanism/Tank Gunnery (THM/TG) unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, General Specification Governing the Manufacture and Inspection of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6920

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MIL-E-70694	- Electronic Control Unit.
MIL-R-70834	- Receiver, Channel X: 11784801.
MIL-T-70835	- Transmitter, Radio: 11784701.

STANDARDS

MILITARY

MIL-STD-109	- Quality Assurance Terms and Definitions.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-2000	- Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA, 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

11784501	- Automatic Tank Target.

SPECIAL PACKAGING INSTRUCTIONS

PS11784501	- Special Packaging Instruction for Automatic Tank Target.

(Copies of drawings, publications and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General description. The automatic tank target assembly consists of an electrical and hydraulic unit which rotates a drive shaft through an angular displacement of 90 degrees that raises and lowers an armored vehicle target. An electronic control unit provides on site front panel controls for on/off mode operations and hit sensitivity selection. The electronic control unit operates the target mechanism on site or remotely through the radio receiver and controls the hit sensor sensitivity, target response to hits, visual hit indicator lamp (VHIL) and gunfire simulator (GUFS). Power is provided by a 12 volt lead acid battery with two power cable connections through the battery box assembly to the automatic tank target assembly.

3.2 Fabrication. The automatic tank target assembly shall be manufactured in accordance with drawing 11784501 and Military Specifications MIL-F-13926 and MIL-STD-454. The electronic control unit shall comply with Military Specification MIL-E-70694. The channel X receiver shall comply with Military Specification MIL-R-70834 and receive radio commands from transmitter per Military Specification MIL-T-70835.

3.2.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2.2 Components and assemblies. The components and assemblies shall comply with all requirements specified in this specification and on drawing 11784501, and all associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Functions.

3.3.1 Commands. The automatic tank target assembly shall, through its electronic control unit and radio receiver, execute commands to raise and lower targets and activate the VHIL and GUFS.

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3.3.2 Mechanism control. The mechanism shall be remotely controlled and shall respond as follows:

a. Upon receiving a remote raise or lower command signal the mechanism shall move the target to the position commanded. The mechanism shall not respond in any manner if commanded to a position that it is already in.

b. The mode of operation shall be selectable directly on the electronic control unit by a rotary switch. This rotary switch shall serve as an on/off switch. At position 0 all electrical power shall be removed from the electronic control unit. At position 1, the target shall be raised and lowered by remote command per 3.3.2.c and hits on the target shall have no effect. At position 2, the target shall be raised and lowered by remote command per 3.3.2.c and a hit on the target when raised will lower target and it will stay lowered until another raise remote signal is received. At position 3, the target shall be raised and lowered by remote command per 3.3.2.c and a hit on the target will lower the target, delay 11 $\pm$ 2 seconds, and then automatically raise the target until another hit occurs and causes it to repeat the above operation. At positions 4, 5 and 6 the VHIL will illuminate when the target is hit. Modes 4, 5 and 6 are the same as modes 1, 2 and 3 respectively except for VHIL operation.

c. Remote signal to raise or lower the target shall be a 12.0 volts $\pm$ 1 volt dc remote pulse equal to or greater than 40 ms in duration. Target shall be raised when signal is applied to plug P1, contact D and target shall be lowered when signal is applied to plug P1, contact F.

d. A hit sensor signal of 12 volts indicating a hit has occurred shall be received through the electronic control unit connectors as follows:

Hit Sensor	Plug	Connector Pin
1	ST-5	A and B
2	ST-6	A and B
3	ST-7	A and B
4	ST-8	A and B

3.3.3 Gunfire simulator control (GUFS). The target device shall operate the gunfire simulator and fire a flash or smoke charge when a 9.3-13.6 Vdc pulse (50 msec duration minimum) signal is received on pin I, J or N on the GUFS input connector indicating a target hit has occurred in any rotary switch position from 1 to 6.

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3.4 Performance requirements.

3.4.1 Output drive.

3.4.1.1 Position. The automatic tank target assembly during operation shall raise a target in accordance with 3.4.1.3 ninety plus or minus five degrees (90 ± 5 degrees) from the horizontal position to the vertical position.

3.4.1.2 Cycling time. The automatic tank target assembly during operation shall raise a target in accordance with 3.4.1.3 in less than eight seconds and lower a target within eight seconds, maximum.

3.4.1.3 Target size and weight. The automatic tank target assembly shall be capable of raising and lowering a target 8.0 ± 0.1 feet high by 12.0 ± 0.1 feet wide by 1.0 ± 0.1 inches thick and weighing 350 ± 10% pounds maximum.

3.4.2 Hydraulic requirements.

3.4.2.1 Oil tank capacity. The automatic tank target hydraulic assembly shall have an oil tank capacity of 2.8 ± 0.1 pints.

3.4.2.2 Oil pump/motor current. The automatic tank target hydraulic oil pump/motor current draw shall not exceed 150 amps.

3.4.2.3 Oil pump pressure. The automatic tank target hydraulic operational oil pump pressure shall not exceed 2000 psi.

3.4.2.4 Inlet oil pressure. The automatic tank target hydraulic inlet oil pressure shall not exceed 2000 psi.

3.4.3 Battery requirements.

3.4.3.1 Battery voltage. The automatic tank target battery voltage shall be 12.0 volts dc nominal and shall not be less than 10.5 volts dc nor more than 15.0 volts dc during operation.

3.5 Environmental requirements.

3.5.1 Temperature.

3.5.1.1 Operating. The automatic tank target assembly shall meet the requirements of 3.3 and 3.4 when stabilized thermally at -25 degrees F and +125 degrees F.

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3.5.1.2 Non-operating (storage). The automatic tank target assembly when subjected to at least four hours at each temperature extreme of -40 degrees F and +160 degrees F shall meet the requirements of 3.3 and 3.4 when returned to and thermally stabilized at ambient temperature (+60 degrees F to +90 degrees F).

3.5.2 Vibration. The automatic tank target assembly shall meet the requirements of 3.3 and 3.4 after being subjected to Method 514, Procedure I-3.3.1, using figure 514.4-1 of MIL-STD-810 for one hour of random vibration in the vertical axis.

3.5.3 Shock. The automatic tank target assembly shall meet the requirements of 3.3 and 3.4 after being subjected to mechanical shock in accordance with Method 516 of MIL-STD-810, Procedure IV, Transit drop test.

3.5.4 Upweight. The automatic tank target assembly shall meet the requirements of 3.3 and 3.4 when weights are used to simulate a continuous 25 MPH horizontal wind directed perpendicular to the target board.

3.5.5 Immersion (non-operating). The immersion test shall be in accordance with 4.7.3.6. If penetration occurs, the applicable item shall be reimmersed and a negative pressure equivalent to three head feet of water shall be applied. The item shall be observed for escaping air bubbles to determine penetration point.

3.6 First article. When specified in the contract or purchase order, sample shall be subjected to first article inspection in accordance with the technical provisions herein. (See 4.3)

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (TASK) F paragraph 1.2 of MIL-STD-2000.

3.8 Workmanship. The workmanship requirements of MIL-STD-454 and MIL-F-13926 shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this

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specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	$+60^{\circ}$ to $+90^{\circ}$F
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of (3) three Tank target assemblies for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in Tables I and II herein and to any or all requirements of the applicable drawings.

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4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogenous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Alternative inspection procedures, methods or equipment, such as statistical process control, tool control, other types of sampling procedures, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the procuring contracting officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the specified quality assurance provisions herein. In cases of dispute as to whether the contractor's proposed alternative(s) provide equal assurance, the provisions of this

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specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program plan or detailed inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

TABLE I. Functional tests.
(100% Inspection)

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Function	3.3	4.8
102. Performance	3.4	4.7.4
103. Soldering	3.7	Visual
104. Workmanship	3.8	4.9
Minor: None		N/A

4.5 Special sampling. Three (3) Automatic Tank Target Assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall have met the requirements and tests in Table I and shall then meet the requirements and tests in Table II. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

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TABLE II. Special sampling

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
105. Automatic tank target assembly	3.4	4.7.4.1
106. Remote control operation	3.3.2	4.8.3
107. Bad hit sensors	3.3.2	4.8.3.5 thru 4.8.3.18
108. Vibration (non-operating)	3.5.2	4.7.3.3
109. Temperature (operating)	3.5.1.1	4.7.2, 4.7.2.2
110. Shock	3.5.3	4.7.3.4
111. Upweighted test	3.5.4	4.7.3.5
112. Immersion test	3.5.6	4.7.3.6

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that

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the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.4.

4.7 Methods of inspection.

4.7.1 Environment. Environment tests shall be conducted in accordance with the general requirements and test methods of MIL-STD-810 unless otherwise specified herein. Testing equipment used in environmental tests shall be in accordance with "Test Facilities and Apparatus" requirements of MIL-STD-810.

4.7.2 Operating.

4.7.2.1 Low temperature. Test shall be performed as specified in MIL-STD-810, Method 502, Procedure II, stabilized at -25 degrees F. While stabilized at -25 degrees F, the assembly shall be operationally checked as specified in 4.7.4 and 4.8.3 to determine compliance with 3.3 and 3.4.

4.7.2.2 High temperature. Test shall be performed as specified in MIL-STD-810, Method 501, Procedure II, stabilized at +125 degrees F. While stabilized at +125 degrees F the assembly shall be operationally checked as specified in 4.7.4 and 4.8.3 to determine compliance with 3.3 and 3.4.

4.7.3 Non-operating.

4.7.3.1 Low temperature. Test shall be performed as specified in MIL-STD-810, Method 502, Procedure I. The assembly shall be stabilized at -40 degrees F for a minimum of 4 hours. After temperature exposure the assembly shall be stabilized at ambient (60 to 90 degrees F) and operationally checked as specified in 4.7.4 and 4.8.3 to determine compliance with 3.3 and 3.4.

4.7.3.2 High temperature. Test shall be performed as specified in MIL-STD-810, Method 501, Procedure I. The assembly shall be stabilized at +160 degrees F for a minimum of 4 hours. After temperature exposure the assembly shall be stabilized at ambient (60 to 90 degrees F) and operationally checked as specified in 4.7.4 and 4.8.3 to determine compliance with 3.3 and 3.4.

4.7.3.3 Vibration. Test shall be performed as specified in MIL-STD-810, Method 514, Procedure I, using figure 514.4-1. Test for one hour along the vertical axis. After vibration, the assembly shall be operationally checked as specified in 4.7.4 and 4.8.3 to determine compliance with 3.3 and 3.4.

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4.7.3.4 Shock. Test shall be performed as specified in MIL-STD-810, Method 516 Procedure IV, Transit Drop test. After the shock impulses, the assembly shall be operationally checked as specified in 4.7.4 and 4.8.3 to determine compliance with 3.3 and 3.4.

4.7.3.5 Upweight. Test shall be performed with weights used to simulate a continuous 25 MPH horizontal wind force directed perpendicular to the target board. The target holding mechanism assembly shall then be operationally checked as specified in 4.7.4 and 4.8.3 to determine compliance with 3.3 and 3.4. Stabilizer bars are to be out to prevent the target from tipping.

4.7.3.6 Immersion test. The following items shall be capable of withstanding the immersion test as specified in MIL-STD-810, Method 512, without being damaged or adversely affected: Hydraulic Assembly 9375758, Hit Sensor 11784450. The ECU shall be tested per military specification MIL-E-70694. All cables and hydraulic lines should be connected to the hydraulic unit with the exception of the battery cables. The applicable items shall be immersed in water and subjected to a positive pressure equivalent to three head feet of water for a 2 hour period + 5 minutes. Subsequent to the immersion test these items shall have no evidence of water penetration.

4.7.4 Performance tests.

4.7.4.1 Target size and weight. The automatic tank target assembly shall be interfaced with the appropriate simulations as in 4.8 and figure 1.

4.7.4.2 Output drive and hydraulic tests. System shall perform as specified in the procedures below utilizing target per 3.4.1.3 to verify the following requirements. Should unit fail to pass any of the following requirements, that unit shall be rejected.

4.7.4.2.1 Position. Measure 90 degree angle using standard measuring equipment to determine compliance with requirement 3.4.1.1.

4.7.4.2.2 Cycling time. Measure raising time using standard measuring equipment to determine compliance with requirement 3.4.1.2.

4.7.4.2.3 Cycling time. Measure raising time using standard measuring equipment to determine compliance with requirement 3.4.1.2.

4.7.2.4 Oil tank capacity. Measure tank capacity using standard measuring equipment to determine compliance with requirement 3.4.2.1.

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4.7.4.2.5 Oil pump/motor current. Measure oil pump/motor current using standard measuring equipment to determine compliance with requirement 3.4.2.2.

4.7.4.2.6 Oil pump pressure. Measure oil pump pressure using standard measuring equipment to determine compliance with requirement 3.4.2.3.

4.7.4.2.7 Inlet oil pressure. Measure inlet oil pressure using standard measuring equipment to determine compliance with requirement 3.4.2.4.

4.7.4.2.8 Battery voltage. Measure battery voltage using standard measuring equipment to determine compliance with requirement 3.4.3.1.

4.8 Function control test procedure.

4.8.1 Equipment. To be designed in accordance with 4.4 and to comply with 3.3, 4.8.3. and Figure 1.

a. Target Load Simulator (TLS)
b. Power Supply (PS)
c. Target Arm Simulator (TAS)
d. GUFS Simulator (GS)
e. Smoke Scorer Simulator (SSS)

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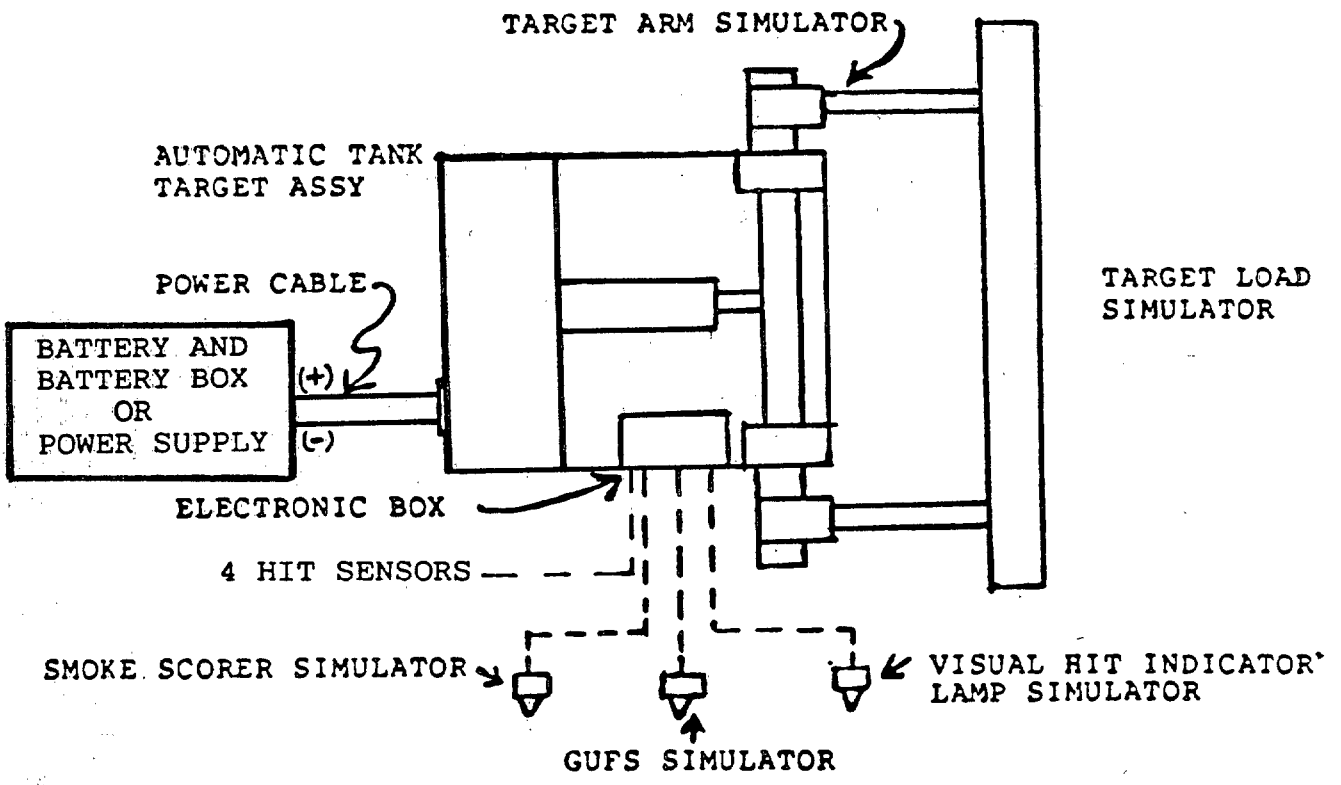


Figure 1. THM/TG overall assembly.

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4.8.2 Simulated test set up. Test set up to simulate the following functions in 4.8.3.

4.8.3 Procedure.

4.8.3.1 Power supply. The BLACK and RED power cables should plug into the 12V dc lead acid battery supply.

4.8.3.2 Position zero. Electronic box function switch should be on the "0" (zero) position.

4.8.3.3 Position one. Put the electronic box function switch on position one.

a. Operate the electronic box test switch momentarily to the up position. Observe that the target simulator raises to the up position and stops within ± 5 degrees of the vertical up position (90 degrees).

b. Tap each of the four hit sensors. Observe that the smoke scorer simulator lamp lights momentarily for each tap. The target simulator should remain in the up position.

c. Operate the electronic box test switch momentarily to the down position. Observe that the target simulator lowers to the down position.

d. Operate the radio transmitter and observe that the target simulator raises and lowers per 3.3.2.c.

4.8.3.4 Position two. Put the electronic box function switch on #2 position.

a. Operate the electronic box test switch momentarily to the up position. Observe that the target simulator raises to the up position and stops within ± 5 degrees of the up position (90 degrees).

b. Tap one of the four hit sensors. Observe that the smoke scorer simulator lamp lights momentarily. The target simulator should lower to the down position.

c. Operate the radio transmitter and observe that the target simulator raises and lowers per 3.3.2.c.

4.8.3.5 Position three. Put the electronic box function switch on #3 position.

a. Operate the electronic box test switch momentarily to the up position. Observe that the target simulator raises to the up position and stops within ± 5 degrees of the up position (90 degrees).

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b. Tap one of the four hit sensors. Observe that the smoke scorer simulator lights momentarily. The target simulator should lower to the down position, delay 9 to 13 seconds, then raise to the up position.

c. Operate the electronic box test switch momentarily to the down position. Observe that the target simulator lowers to the down position.

d. Operate the radio transmitter and observe that the target simulator raises and lowers per 3.3.2.c.

4.8.3.6 Position four. Put the electronic box function switch on position #4.

a. Operate the electronic box test switch momentarily to the up position. Observe that the target simulator raises to the up position and stops within ± 5 degrees of the up position (90 degrees).

b. Tap each of the four hit sensors. Observe that the smoke scorer simulator lamp lights momentarily for each tap. The visual hit indicator lamp simulator lights momentarily. The target simulators should remain in the up position.

c. Operate the electronic box test switch momentarily to the down position. Observe that the target simulator lowers to the down position.

d. Operate the radio transmitter and observe that the target simulator raises and lowers per 3.3.2.c.

4.8.3.7 Position five. Put the electronic box function switch on position #5.

a. Operate the electronic box test switch momentarily to the up position. Observe that the target simulator raises to the up position and stops within ± 5 degrees of the up position (90 degrees).

b. Tap one of the four hit sensors. Observe that the smoke scorer simulator lamp and visual hit indicator simulator lamp light momentarily. The target simulator should lower to the down position.

c. Operate the radio transmitter and observe that the target simulator raises and lowers per 3.3.2.c.

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4.8.3.8 Position six. Put the electronic box function switch on position #6.

a. Operate the electronic box test switch momentarily to the up position. Observe that the target simulator raises to the up position and stops within ± 5 degrees of the up position (90 degrees).

b. Tap one of the four hit sensors. Observe that the smoke scorer simulator lamp and visual hit indicator simulator lamp light momentarily. The target simulator should lower to the down position, delay 9 to 13 seconds, then raise to the up position.

c. Operate the electronic box test switch momentarily to the down position. Observe that the target simulator lowers to the down position.

d. Operate the radio transmitter and observe that the target simulator raises and lowers per 3.3.2.c.

4.9 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in 3.7.

5. PACKAGING

5.1 General. Preservation, packaging, packing and marking shall be in accordance with special packaging instruction PS11784501. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification details the specific requirements for design, performance, quality assurance, and testing for the automatic tank target assembly. The tank target assembly will be used as part of the Tank Holding Mechanism/Tank Gunnery (THM/TG) and will support live fire armor weapons training and qualification programs. These systems will provide training in identifying and firing on simulated stationary armored vehicles during training exercises.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Place of inspection if not at place of manufacture.

d. Instructions for shipment of production lot samples.

e. Provision for preparation for delivery if required for indefinite storage of inter-plant shipment.

f. Provision for the supply, maintenance, and disposition of test equipment for acceptance inspection process.

g. Procuring activity designation assembly number(s) and nomenclature(s).

h. Serialization requirements, if applicable.

i. Packaging requirements, if other than specified in Section 5.

j. Certificates of conformance for each lot or shipment of product.

k. Issue of DODISS to be cited in solicitation and if required, the specific issue of individual document to reference (see 2.1.1).

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of design as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I (D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

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6.5 Definitions.

6.5.1 Trainers. Training device terms are defined in MIL-HDBK-220. Other definitions, if required, will be furnished upon written request to the Procuring Contracting Officer.

6.5.2 Contractor. Unless otherwise specified, the word Contractor used throughout this specification shall mean any supplier or producer of items or material to the Government. This shall include commercial contractors, sub-contractors, Government Owned-Contractor Operated (GOCO), and Government Owned-Government Operated (GOGO) plants.

6.5.3 Technical data package. A technical description of an item adequate for use in procurement. The description defines the required design configuration and assures adequacy of item performance. It consists of all applicable technical data such as plans, drawings, and associated lists, specifications, standards, models, performance requirements, quality assurance provisions and packaging data.

6.6 Subject term (keyword) listing.

a. Target Holding Mechanism/Tank Gunnery (THM/TG).
b. Gunfire Simulator (GUFS).
c. Visual Hit Indicator Lamp (VHIL).

6.7 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 6920-A015)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-A-70833A (AR)	4 September 1991

3. DOCUMENT TITLE: AUTOMATIC TANK TARGET: 11784501

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Commander US Army ARDEC	(1) Commercial (201) 724-6674	(2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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